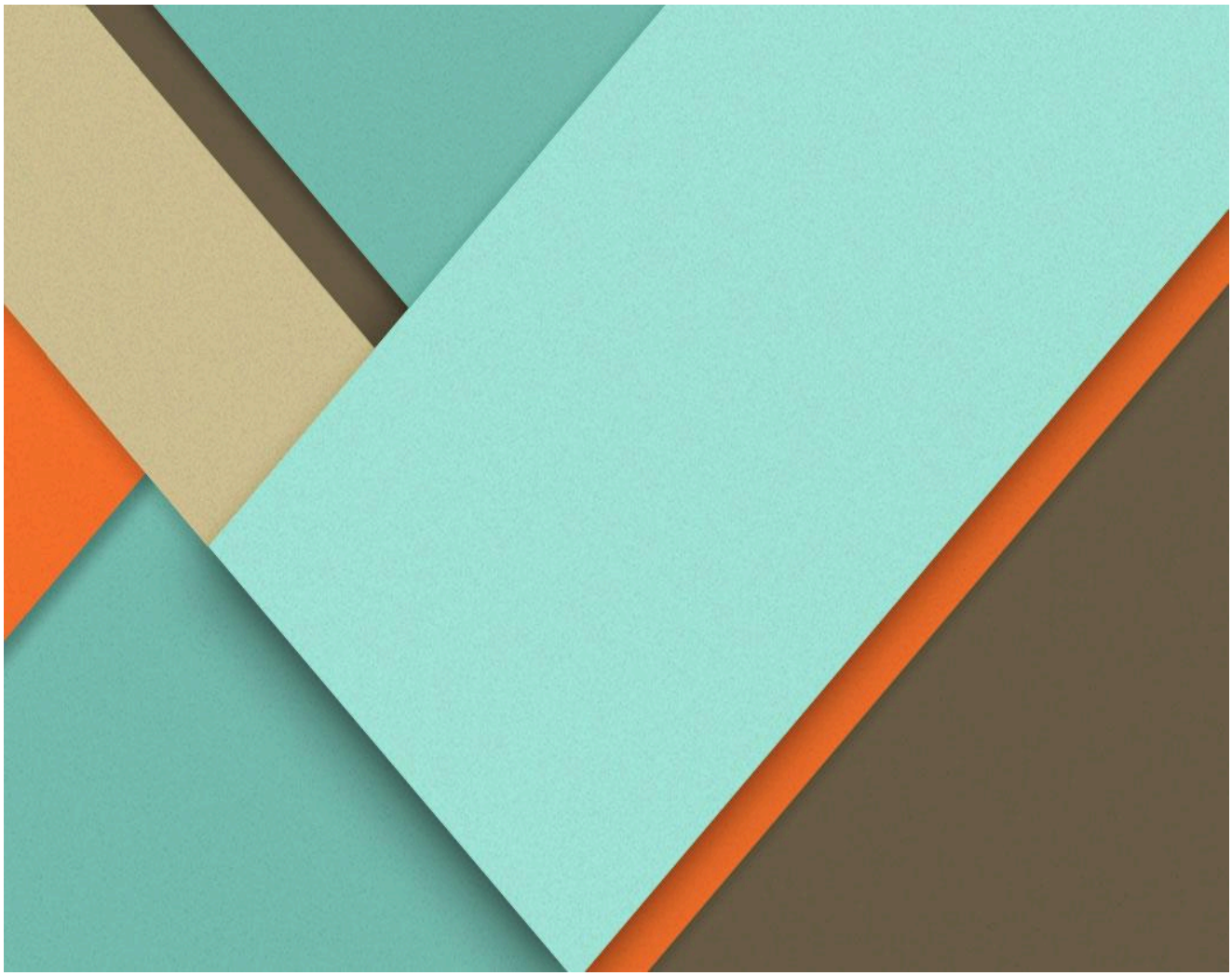

Employee Salary Analysis System

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BATCH / PROGRAM: B-Tech CSE
DATA FUNDAMENTALS COURSE

TITLE : Employee Salary Analysis System

PROBLEM STATEMENT

Organizations need to analyze employee salary data to understand salary distribution, department-wise pay differences, and the impact of experience on salary. Manual analysis is time-consuming and inefficient.

This project aims to analyze employee data using SQL, Excel, and Python to generate meaningful insights that help in decision-making related to employee compensation.

SOLUTION / FEATURES

The project provides a complete analysis of employee salary data using multiple tools:

- SQL is used to store and query employee data efficiently.
- Excel is used for sorting, pivot tables, and quick visualization.
- Python (Pandas & Matplotlib) is used for data analysis and graphical representation.

Key Features:

- Calculation of average salary
- Identification of highest-paid employees
- Department-wise salary comparison
- Experience vs salary analysis
- Visual representation using charts

TECH STACK

- SQL (Oracle SQL*Plus) – Data storage and querying
- Microsoft Excel – Data cleaning and pivot analysis
- Python (Pandas, Matplotlib) – Data analysis and visualization

INSIGHTS (IMPORTANT)

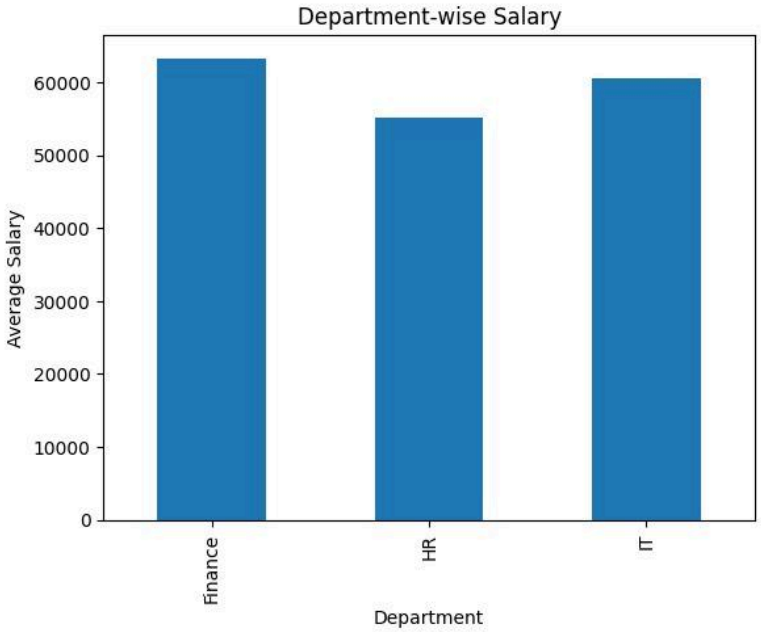
- Employees with higher experience generally earn higher salaries, indicating a positive relationship between experience and pay.
- The Finance department has the highest average salary among all departments.
- The HR department shows moderate salary levels compared to IT and Finance.
- The highest-paid employees have more than 8 years of experience.
- Salary distribution shows variation across departments and experience levels.

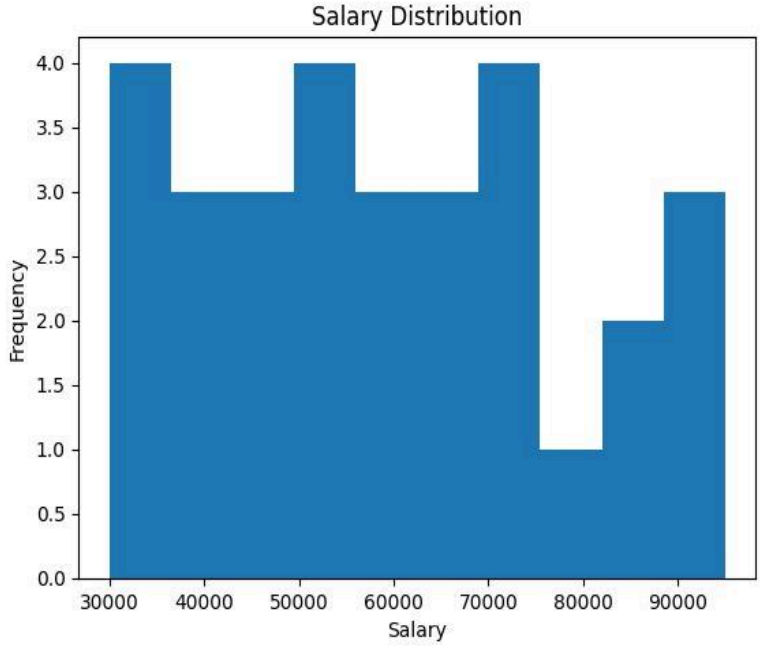
Excell

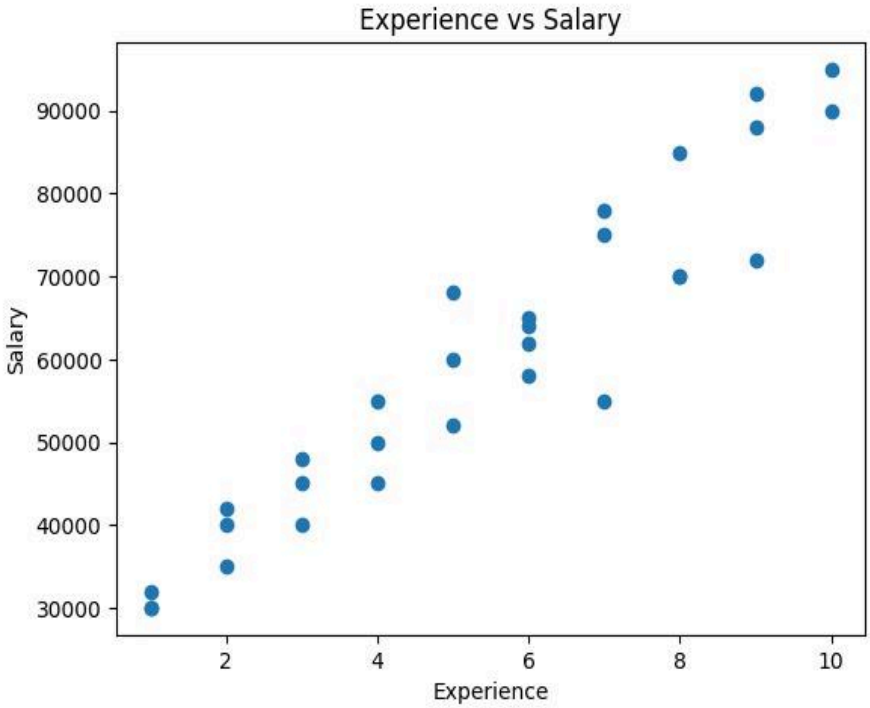
salary
100000
80000
60000
40000
20000
0
0
2
4
6
8
10
12

Average of salary
25%
27%
25%
23%
Finance
HR
IT
Grand Total

Python


Department-wise Salary
Average Salary
0
10000
20000
30000
40000
50000
60000
Finance
HR
IT
Department


Salary Distribution
Frequency
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
30000
40000
50000
60000
70000
80000
90000
Salary


Experience vs Salary
Salary
30000
40000
50000
60000
70000
80000
90000
2
4
6
8
10
Experience

UNIQUE POINTS

- Integration of multiple tools (SQL, Excel, Python) in one project
- Use of real-world employee dataset
- Automated analysis instead of manual calculations
- Clear visualization of salary trends and patterns

FUTURE IMPROVEMENTS

- Integration with real-time company databases
- Development of a web dashboard for interactive analysis
- Inclusion of more factors like performance ratings and bonuses
- Use of advanced analytics and machine learning for salary prediction